Managing for Quality and Performance Excellence, 9th Edition

Summary of Key Points and Terminology

Chapter 2 – Foundations of Quality Management

- W. Edwards Deming, Joseph Juran, and Philip Crosby are recognized as the most influential thinkers and leaders of modern quality management. A. V. Feigenbaum and Kaoru Ishikawa have also made significant contributions to modern quality management practices.
- Deming's philosophy is based on improving products and services by reducing uncertainty and variation in design, manufacturing, and service processes, driven by the leadership of top management. The Deming Chain Reaction suggests that quality improvement leads to long-term business and economic prosperity. The **14 Points** represent the practices that Deming advocated for achieving quality excellence.
- Deming synthesized the underlying foundations of the 14 Points into four simple elements that he called a **System of Profound Knowledge**:
    1. Appreciation for a system
    2. Understanding variation
    3. Theory of knowledge
    4. Psychology
- A **system** is a set of functions or activities within an organization that work together for the aim of the organization. Systems thinking is critical in applying quality principles because the organizational linkages among various functions of an organization must be in alignment to meet the needs of customers and other stakeholders.
- Variation exists in any production and service process, generally due to factors inherent in the design of the system, which cannot easily be controlled. Excessive variation results in products that fail or perform erratically and inconsistent service that does not meet customers' expectations. Statistical methods are the

primary tools used to identify and quantify variation. Deming suggested that management first understand, and then work to reduce variation through improvements in technology, process design, and training.

- The **theory of knowledge** is the branch of philosophy concerned with the nature and scope of knowledge, its presuppositions and basis, and the general reliability of claims to knowledge. Basically, managers need to understand how things work and why decisions that affect the future should be effective.
- Much of Deming's philosophy is rooted in psychology and based on understanding human behavior and treating people fairly. People can be motivated intrinsically and extrinsically; however, the most powerful motivators are intrinsic.
- Joseph Juran's philosophy seeks to provide change within the current American management system. Juran defined quality as *fitness for use* – the user-based definition we described in Chapter 1. His **Quality Trilogy**—planning, control, and improvement—provides a direction for quality assurance in organizations.
- Juran specified a detailed program for quality improvement, known as the **breakthrough sequence**, consisting of proof of the need, project identification, organization for breakthrough, the diagnostic journey, the remedial journey, and holding the gains.
- Philip Crosby's approach to quality is summarized in his **Absolutes of Quality Management** – conformance to requirements, no such thing as a quality problem, doing the job right the first time, cost of quality measurement, and zero defects as the only performance standard – and **Basic Elements of Improvement** – determination, education, and implementation. He places more emphasis on behavioral change rather than on the use of statistical techniques as advocated by Deming and Juran.
- A. V. Feigenbaum views quality as a strategic business tool and coined the phrase "total quality control." He promoted the importance of shifting quality responsibility to everyone in an organization and developing cost of quality approaches. His **Three Steps to Quality** are quality leadership, modern quality technology, and organizational commitment.

- Kaoru Ishikawa was instrumental in the Japanese quality movement, particularly in advocating a company-wide quality control approach, the use of employee teams, and the use of problem-solving tools for quality improvement.
- Total quality can be characterized by its principles, practices, and techniques. **Principles** are the foundation of the philosophy, **practices** are activities by which the principles are implemented, and **techniques** are tools and approaches that help managers and workers make the practices effective. All are vital for achieving high quality and performance excellence.
- The three core principles of TQ are customer focus, teamwork, and continuous improvement. These principles have evolved and are now summarized by the quality management principles that define the ISO 9000 standards (see Table 2.2).
- Quality management practices represent the approaches that organizations use to achieve TQ principles (see Table 2.3).
- Quality management techniques include a wide variety of tools to plan work activities, collect data, analyze results, monitor progress, and solve problems.
- Continuous improvement refers to both incremental--small and gradual--and breakthrough--large and rapid--improvement.
- **Statistical thinking** is a philosophy of learning based on principles of understanding that all work occurs in a system of interconnected processes, variation exists in all processes, and variation must be understood and reduced.
- **Common causes of variation** are inherent to a process, generally account for most observed variation, and cannot be identified on an individual basis or controlled. **Special (assignable) causes of variation** are sporadic in nature and result from external disturbances that can usually be identified statistically and either explained or corrected. A system governed only by common causes is called a stable system.
- Management can make two fundamental mistakes in attempting to improve a process: 1. To treat as a special cause any fault, complaint, mistake, breakdown, accident, or shortage when it actually is due to common causes. 2. To attribute to

common causes any fault, complaint, mistake, breakdown, accident, or shortage when it actually is due to a special cause.

- Not understanding the differences between common and special causes can result in increasing the variation through tampering with stable systems, or missing opportunities to reduce special cause variation when it exists. Deming's **Red Bead experiment** and **Funnel experiment** can help clarify the differences between common and special causes and improve managers' abilities to make effective decisions.
- The lessons from the Red Bead experiment are that variation exists in systems and, if stable, can be predicted; all the variation in the production of red beads, and the variation from day to day of any Willing Worker, came entirely from the process itself; numerical goals are often meaningless; and management is responsible for the system.
- According to the ASQ online glossary, a **quality management system (QMS)** can be considered a mechanism for managing and continuously improving core processes to "achieve maximum customer satisfaction at the lowest overall cost to the organization." A quality management system represents a specific implementation of quality concepts, standards, methods and tools, and is unique to an organization. A QMS provides a basis for documenting processes used to control and improve operations.
- A **quality policy** is a formal document that demonstrates a commitment to achieving high quality and meeting customer expectations. A **quality manual** serves as a permanent reference for implementing and maintaining the system. **Internal audits** focus on identifying whether documented procedures are being followed and are effective, and reporting the issues to management for corrective action.
- The **ISO 9000 family of standards** are a set of standards and guidelines for quality management systems that represents an international consensus on good quality management practices. They provide a comprehensive framework for designing and managing a quality management system and help organizations

establish a process orientation and the discipline to document and control key processes.

- An effective QMS needs to be integrated with enterprise systems such as ERP, MES, and SCM, and should focus on actionable decision making, seeking the root causes of problems, and improving processes and systems. It should drive the principles of quality management throughout the organization by fostering effective practices to implement the principles.